

2023 TRIAL EXAMINATION

Mathematics Advanced

**General
Instructions**

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

**Total Marks:
100**

Section I – 10 marks (pages 2–5)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 6–23)

- Attempt Questions 11–28
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Which of the following is equivalent to $\frac{d}{dx} \log_2(x - 5)$?

A. $\frac{1}{x \ln 2}$

B. $\frac{1}{(x - 5) \ln 2}$

C. $\frac{2}{x}$

D. $\frac{2}{x - 5}$

2 The first four terms of a sequence are 3, 8, 13, 18.

What is the recursive definition of this sequence?

A. $T_n = T_{n-1} + 5$

B. $T_n = T_{n+1} + 5$

C. $T_n = 2T_{n-1} + 2$

D. $T_n = 2T_{n+1} + 2$

3 Which of the following statements is true regarding data which is normally distributed?

A. The mean cannot be zero.

B. The standard deviation cannot be zero.

C. The distribution may be skewed.

D. The standard deviation can be negative.

- 4 A graph, $g(x)$, is obtained by transforming $f(x)$ such that $g(x) = -\frac{1}{2}f(x - 1) + 2$.

The point $(3, 4)$ is on the graph of $f(x)$. Where does this point correspond to on $g(x)$?

- A. $(4, 0)$
 - B. $(2, 0)$
 - C. $(-2, 6)$
 - D. $(-1, 6)$
- 5 At the beginning of a month, \$40,000 is deposited into an account earning interest at 18% p.a., compounded monthly. At the end of each month, \$2000 is withdrawn from the account. If the balance falls below \$2000, the remaining amount is withdrawn.

Which of the following is closest to the amount of time until the account reaches a balance of \$0?

You may use the table below, which gives the present value of an annuity of \$1.

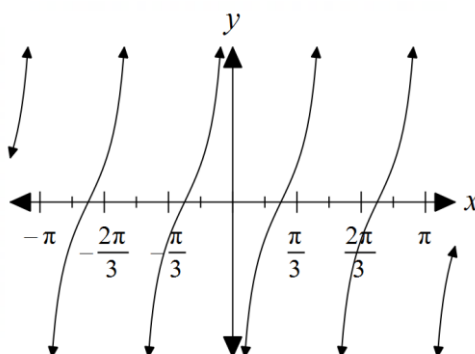
Present Value of an Annuity of \$1						
Periods	Interest Rate per period					
	0.25%	0.50%	0.75%	1.00%	1.25%	1.50%
1	0.9975	0.9950	0.9926	0.9901	0.9877	0.9852
6	5.9478	5.8964	5.8456	5.7955	5.7460	5.6972
12	11.8073	11.6189	11.4349	11.2551	11.0793	10.9075
18	17.5795	17.1728	16.7792	16.3983	16.0295	15.6726
24	23.2660	22.5629	21.8891	21.2434	20.6242	20.0304
30	28.8679	27.7941	26.7751	25.8077	24.8889	24.0158

- A. 1 year
- B. 1 year and 10 months
- C. 2 years
- D. 2 years and 6 months

6 If $\log_b 3 = x$ and $\log_b 4 = y$, which expression is equivalent to $\log_b(36b)$?

- A. $2xy$
- B. $2xyb$
- C. $2x + y + 1$
- D. $x^2 + y + b$

7 Which of the following functions best represents the graph below?



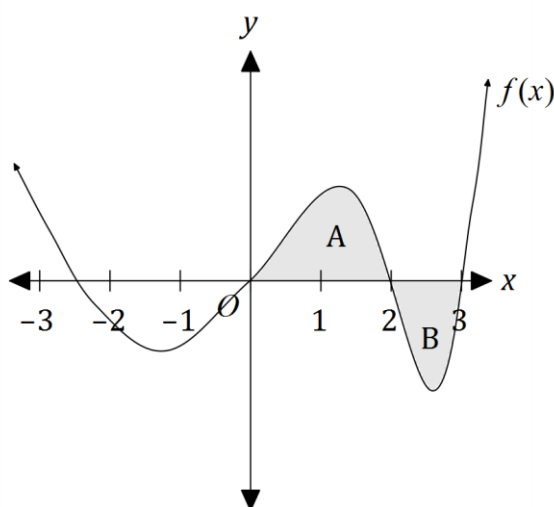
- A. $y = \tan\left(4x + \frac{\pi}{2}\right)$
- B. $y = \tan\left(4x + \frac{\pi}{4}\right)$
- C. $y = \tan\left(2x + \frac{\pi}{2}\right)$
- D. $y = \tan\left(2x + \frac{\pi}{4}\right)$

8 A website claims to be 98% accurate at detecting plagiarism. That is, it will return the correct result for a paper, “Plagiarised” or “Not Plagiarised,” 98% of the time.

It is known that 5% of submitted papers at a school are plagiarised. If a teacher checks a randomly selected paper using the website and gets the result “Plagiarised,” what value is closest to the probability that it is a plagiarised paper?

- A. 69%
- B. 72%
- C. 95%
- D. 98%

- 9 The graph of a function $f(x)$ is shown below, where the shaded area labelled A is $\frac{3}{2}$ of the shaded area labelled B.



If $\int_0^3 f(x)dx = 2$, what is the area of A?

- A. 6
 - B. 4
 - C. $\frac{6}{5}$
 - D. $\frac{4}{5}$
- 10 Let $f(x)$ be an odd, continuous function with domain $(-\infty, \infty)$ and range $(-\infty, \infty)$.

Let $g(x)$ be an even rational function with vertical asymptotes only at $x = -1$ and $x = 1$.

Which of the following MUST be true about $g(f(x))$?

- A. $g(f(x))$ has exactly two horizontal asymptotes
- B. $g(f(x))$ has at least two vertical asymptotes
- C. $g(f(x))$ has exactly two vertical asymptotes
- D. $g(f(x))$ has exactly one vertical asymptote

Mathematics Advanced

Section II Answer Booklet

Section II

90 marks

Attempt Questions 11–28

Allow about 2 hours and 45 minutes for this section

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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Question 11 (1 marks)

Find the values of x which satisfy the equation $x^2 - 6x = -5$

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Question 12 (2 marks)

Rationalise the denominator of $\frac{\sqrt{2}}{1-\sqrt{3}}$ writing your answer in a simplified form.

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Question 13 (2 marks)

Use two applications of the trapezoidal rule to find an approximation of the value of the integral $\int_0^2 e^{-\frac{x^2}{2}} dx$. Give your answer correct to 2 decimal places.

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Question 14 (3 marks)

- (a) Find the sum of the first 80 odd positive integers.

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- (b) The first 3 terms of a series are
- $\frac{1}{2}$
- ,
- $\frac{1}{3}$
- , and
- $\frac{1}{4}$
- .

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Does the series have a limiting sum? If so, find the limiting sum. If not, show why it does not.

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Question 15 (4 marks)

The two-way table shows the percentages of blood types in Australia.

	Positive	Negative
O	40%	9%
A	31%	7%
B	8%	2%
AB	2%	1%

- (a) If 600 Australians are selected randomly to participate in a medical study, how many of them would be expected to have the AB blood type?

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Question 15 continues on page 9

Question 15 (continued)

- (b) If two Australians were chosen at random, what is the probability that one is 0 positive and the other is 0 negative? 3

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Question 16 (5 marks)

Find the following derivatives, simplifying your answers where possible.

- (a) $\frac{d}{dx} e^{2x} \cos(x + 1)$ 2

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- (b) $\frac{d}{dx} \frac{\ln x}{x^2 - 1}$ 3

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Question 17 (4 marks)

(a) Find $\int \frac{x+1}{x^2+2x+1} dx$ **2**

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(b) Evaluate $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sec^2 \frac{x}{2} dx$ **2**

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Questions 11-17 are worth 21 marks in total
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Question 18 (11 marks)

Padma is interested in finding out if there is a relationship between using nicotine and school performance in Year 12 students in Australia by surveying Year 12 students at her school.

- (a) Identify the population and sample in Padma's study.

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- (b) Padma wishes to use systematic sampling to create a sample of 30 students from last year's graduating class of 330 Year 12 students.

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Give a detailed suggestion on how Padma can create her sample using the desired method.

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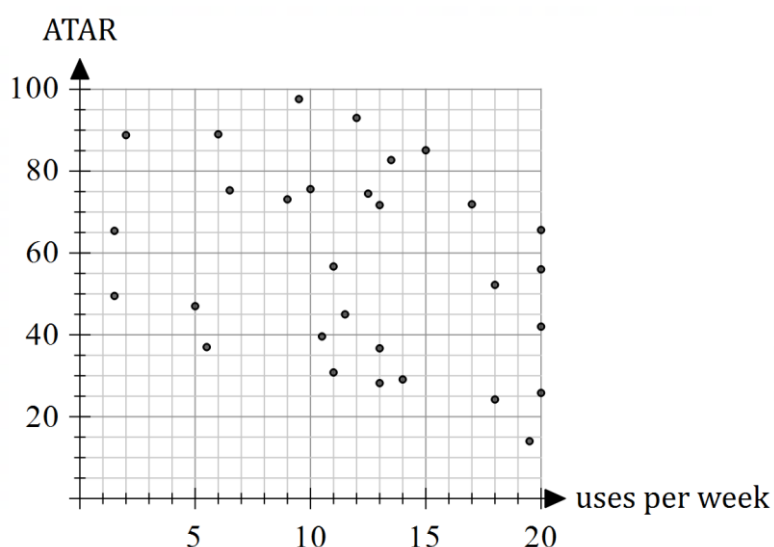
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- (c) After gathering the survey data from her sample, Padma graphs the results on the scatterplot shown below. The horizontal axis represents the number of times the student reported using nicotine per week during Year 12, and the vertical axis represents the student's ATAR.



Question 18 continues on page 12

Question 18 (continued)

- (i) Describe the relationship, including its strength and direction, Padma has found between nicotine use and Year 12 students' ATAR in her sample. 2

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- (ii) According to her statistics software, the least-squares regression line that fits her data is given by $y = 74 - 1.38x$, where x represents the number of times the student reported using nicotine per week during Year 12, and y represents the student's ATAR. 3

Graph this line on the scatterplot on page 11 and predict the ATAR of a student who uses nicotine 16 times per week.

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- (iii) Padma claims that her study shows that nicotine use causes a decrease in Year 12 students' ATAR. 3

Comment on the validity of Padma's claim and give TWO reasons to justify your comment.

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Question 19 (3 marks)

Find the values of x for which $y = x^4 - 2x^2 + x$ is concave up. Give your answer using exact values.

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Question 20 (2 marks)

A die is rolled twice. Let event A be $A = \{\text{the first roll is a 1}\}$ and let event B be $B = \{\text{sum of the two rolls is 7}\}$.

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Show that events A and B are independent.

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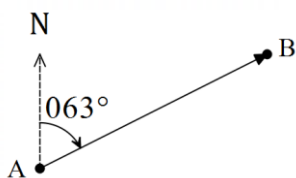
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Questions 11-20 are worth 37 marks in total
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Question 21 (4 marks)

A plane flies from City A on a bearing of 063° and distance of 803 km to City B. It then flies due south to City C. The plane then flies back to City A, which is 720 km from City C. The first leg of the trip is shown in the diagram below.



- (a) Explain why there is not enough information to determine the correct bearing on which the plane must fly to return to City A. **1**

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- (b) Find the possible distance(s) of City C from City B. **3**

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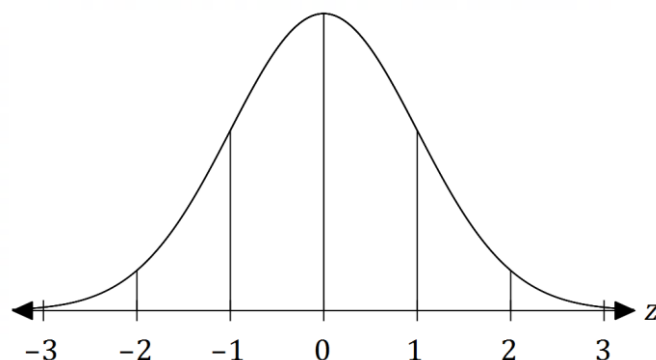
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Question 22 (5 marks)

The weights of newborn babies are normally distributed with a mean of 7.5 pounds and a standard deviation of 1.1 pounds.

- (a) A normal distribution curve is shown below, where the vertical lines represent z-scores from -3 to 3 . **1**



A baby is considered underweight if their birthweight has a z-score of less than -1.8 . Shade the area under the normal distribution curve representing the percentage of babies that are born underweight.

- (b) Below what birthweight, in pounds, is a baby considered to be underweight? **2**
Give your answer to one decimal place.

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- (c) If the area under the normal distribution curve below $z = 1.8$ is equal to 0.9641 , how many babies in $1,000$ would be expected to be born underweight? **2**

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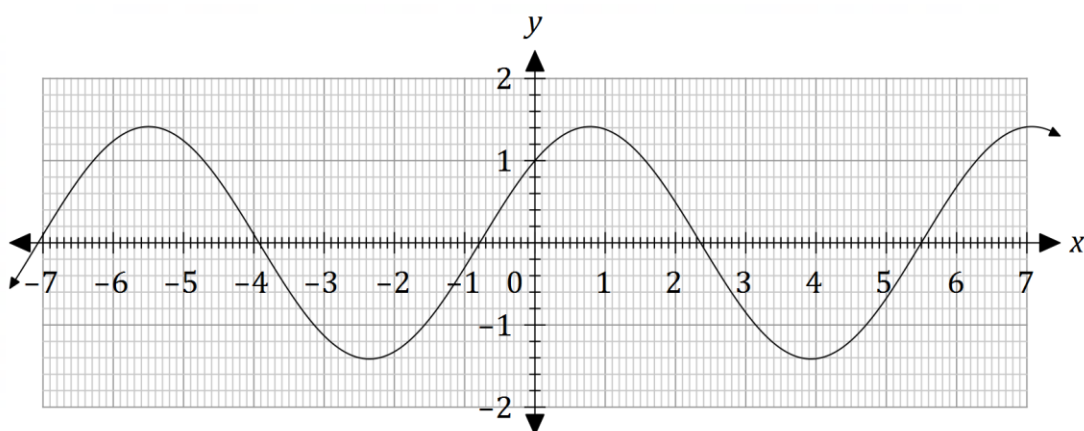
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Question 23 (3 marks)

The graph of $y = \sin(x) + \cos(x)$ is shown on the axes below.



- (a) Mark the approximate solutions to the equation $\sin(x) + \cos(x) = 1$ on the domain $-\pi \leq x \leq \pi$. Write the answer(s) below, approximating your answer(s) to one decimal place.

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- (b) Find the approximate solution(s) to $\sin(x) + \cos(x) + x = 0$ on the domain $-\pi \leq x \leq \pi$. Give your answer to the nearest tenth.

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Question 24 (10 marks)

Consider the function $y = f(x) = \frac{e^x}{x}$.

(a) Explain why $f(x)$ has no x -intercepts.

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(b) Find any stationary points of $f(x)$ and determine their nature.

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(c) Find the concavity of $f(x)$ on each side of its vertical asymptote and show that $f(x)$ has no inflection points.

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Question 24 continues on page 18

Question 24 (continued)

- (d) By considering the sign of $f(x)$, $f'(x)$ and $f''(x)$ on each side of the vertical asymptote, as well as any stationary points, sketch a graph of $f(x)$.

2

Questions 11-24 are worth 59 marks in total

Question 25 (6 marks)

A continuous random variable X has the probability density function

$$f(x) = \begin{cases} kx^2 - 3kx + 4k, & 0 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

where k is a positive constant.

- (a) Find the value of k . **2**

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- (b) Find the mode of random variable X , as well as its least likely outcomes. **2**

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- (c) Given the median of the distribution is approximately 2.77 (do NOT prove this), find $P(X < 3 | X > 2.77)$. **2**

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Question 26 (8 marks)

Jean takes out a reducing-balance home loan of \$580,000. The loan has an interest rate of 4.2% p.a., compounded monthly, and is to be paid over 30 years in equal repayments of M dollars at the end of each month.

- (a) Show that $A_n = 580000 \times 1.0035^n - \frac{M(1.0035^n - 1)}{0.0035}$ represents the amount owing on the loan after n months. 2

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- (b) Find M , the monthly payment, for Jean's home loan. 2

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Question 26 continues on page 21

Question 26 (continued)

- (c) If Jean pays an extra \$100 each month on their loan, find how much interest is saved over the life of the loan. Give your answer to the nearest hundred dollars.

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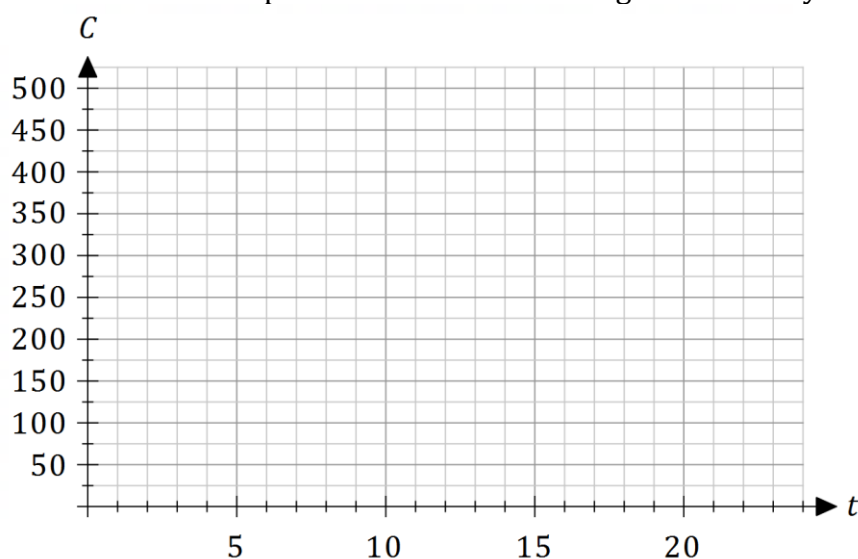
Question 27 (11 marks)

Claire feeds her pet rat at regular intervals and must monitor the rat's insulin levels due to a medical condition. Claire finds the rat's insulin levels follow the shape of a sine curve over a 12-hour period. She has recorded the following information on Monday:

Time	06:00	08:00	09:00	12:00	13:00	15:00	17:00
Insulin Level (pmol/L)	300	450	975	375	250	175	250

- (a) Plot the information in the table on the axes below, where C represents the rat's insulin level and t represents time since midnight on Monday.

1



Question 28 continues on page 22

Question 27 (continued)

- (b) Assuming the rat's measured maximum and minimum insulin levels are its actual maximum and minimum insulin levels, respectively, find $C(t) = A \sin(Bt + C) + D$, where t is the time in hours since midnight on Monday. 4

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- (c) Sketch the graph of $C(t)$ on the axes in part (a) on the previous page for $0 \leq t \leq 24$. The space below may be used for working out, if needed. 1

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- (d) What is the rat's predicted insulin level at 20:00, and what is the rate at which it is changing? Give your answers to the nearest whole number. 2

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- (e) Find the time on Monday after 15:00 that the rat's insulin level will be 200 pmol/L. 3

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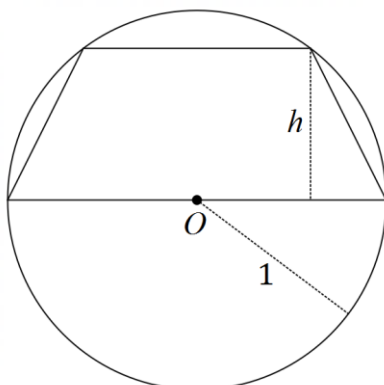
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Question 28 (6 marks)

A trapezium is inscribed in a circle centred at O with a radius of 1, such that one of its bases is a diameter of the circle, as shown in the diagram below. The height of the trapezium, h , may vary.



- (a) Show that the area of the trapezium can be given by $A = h\sqrt{1 - h^2} + h$.

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- (b) Find the exact value of the maximum area of the trapezium.

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Section II extra writing space

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Section II extra writing space

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2023 TRIAL
EXAMINATION

Mathematics Advanced

WORKED SOLUTIONS

Section I

Question	Solution	Answer
1	From the reference sheet: $\frac{dy}{dx} = \frac{f'(x)}{(\ln a)(x)}$ $\therefore = \frac{1}{(\ln 2)(x-5)} \text{ (Option B)}$	B
2	Sequence is produced by adding 5 to the previous term. Can be expressed as: $T_n = T_{n-1} + 5$ (Option A)	A
3	If the standard deviation is zero, all data points would be equal and would not be normally distributed. (Option B)	B
4	$g(x) = -\frac{1}{2}f(x-1) + 2$, hence $f(x)$ has been translated one unit to the right, dilated vertically by a factor of $\frac{1}{2}$ and reflected across the x -axis, and then translated vertically up 1. Thus, the x -coordinate will become $3 + 1 = 4$ and the y -coordinate will become $4 \times -\frac{1}{2} + 2 = 0$. Hence $(3, 4) \rightarrow (4, 0)$ (Option A)	A
5	Interest rate: $\frac{18}{12}\% = 1.5\%$ $40000 = 2000 \times F$ $F = \frac{40000}{2000} = 20$ The factor in the 1.5% column closest to 20 is in the 24-period row. 24 months = 2 years (Option C)	C
6	$\log_b(36b) = \log_b 36 + \log_b b$ $= \log_b 9 + \log_b 4 + \log_b b$ $= \log_b 3^2 + \log_b 4 + \log_b b$ $= 2 \log_b 3 + \log_b 4 + 1$ $= 2x + y + 1$ (Option C)	C
7	Asymptote for $\tan(x)$ typically occurs at $x = \frac{\pi}{2}$. The given graph shows asymptote at $x = 0$, thus graph has been translated horizontally $\frac{\pi}{2}$. $\tan(x)$ typically has two cycles within the domain $[-\pi, \pi]$. Number of cycles has doubled. Thus $y = \tan(2x + \frac{\pi}{2})$ (Option C)	C
8	$P(\text{plagiarised} \text{positive result})$ $= \frac{P(\text{plagiarised} \cap \text{positive result})}{P(\text{positive result})}$ $= \frac{0.05 \times 0.98}{0.05 \times 0.98 + 0.95 \times 0.02}$ (Denominator must include both true positives and false positives.) $= 0.72 = 72\%$ (Option B)	B

9	$A = \frac{3}{2}B,$ $\int_0^3 f(x)dx = 2 = A - B$ (as B is negative) $\therefore \frac{3}{2}B - B = 2$ $\frac{B}{2} = 2$ $B = 4$ $\therefore A = \frac{3}{2} \times B = \frac{3}{2} \times 4 = 6$ (Option A)	A
10	As $f(x)$ is has domain $(-\infty, \infty)$ and range $(-\infty, \infty)$ all inputs for x are valid and all outputs for y will be produced at least once. Thus, the inputs which correspond to vertical asymptotes on $g(x)$ will be outputs of the inner function $f(x)$ at least once, hence the composition will have at least two vertical asymptotes. If $f(x)$ is one-to-one, then the composition will have only two asymptotes; however, if $f(x)$ is many-to-one, the composition may have more than two asymptotes, as more than one input may produce the same output which gives at least one of the vertical asymptotes when input to $g(x)$. (Option B)	B

Section II

Question 11	
Rearrange equation: $x^2 - 6x + 5 = 0$ $(x - 5)(x - 1) = 0$ $x = 5$ or $x = 1$	1
Question 12	
$\frac{\sqrt{2}}{1 - \sqrt{3}} \cdot \frac{1 + \sqrt{3}}{1 + \sqrt{3}}$ $= \frac{\sqrt{2}(1 + \sqrt{3})}{(1 - \sqrt{3})(1 + \sqrt{3})}$ $= \frac{(\sqrt{2} + \sqrt{6})}{1 - 3}$ $= -\frac{\sqrt{2} + \sqrt{6}}{2}$	2
Question 13	
Using trapezoidal rule: $\int_0^2 e^{-\frac{x^2}{2}} = \frac{2 - 0}{2} \left\{ e^{-\frac{(0)^2}{2}} + e^{-\frac{(2)^2}{2}} + 2 \left(e^{-\frac{(1)^2}{2}} \right) \right\}$ $= 1 \left\{ e^0 + e^{-2} + 2e^{-\frac{1}{2}} \right\}$ ≈ 2.35	2

Question 14		
(a)	Arithmetic sum, $a = 1, d = 2, n = 80$ $S = \frac{80}{2}(2 \times 1 + 79 \times 2) = 6400$	1
(b)	It does not have a limiting sum. This is because the sequence is not geometric: $\frac{T_2}{T_1} = \frac{\frac{1}{3}}{\frac{1}{1}} = \frac{2}{3}, \frac{T_3}{T_2} = \frac{\frac{1}{4}}{\frac{2}{3}} = \frac{3}{8}$ $\frac{T_2}{T_1} \neq \frac{T_3}{T_2} \therefore$ the sequence is not geometric.	2

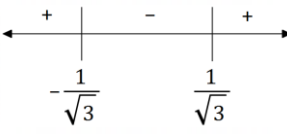
Question 15		
(a)	$600 \times 3\% = 18$	1
(b)	(A tree diagram is not necessary but may be helpful.) $P(O^+ \cap O^-) = 0.4 \times 0.09 + 0.09 \times 0.4 = 7.2\%$	3

Question 16		
(a)	$\frac{dy}{dx} = -\sin(x+1)e^{2x} + 2\cos(x+1)e^{2x}$	2
(b)	$\frac{dy}{dx} = \frac{\frac{x^2-1}{x} - 2x \ln x}{(x^2-1)^2} = \frac{\frac{x^2-1-2x^2 \ln x}{x}}{(x^2-1)^2} = \frac{x^2-1-2x^2 \ln x}{x(x^2-1)^2}$	3

Question 17		
(a)	$\frac{1}{2} \ln(x^2 + 2x + 1) + c$	2
(b)	$\left[2 \tan\left(\frac{x}{2}\right)\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 2 \tan\left(\frac{\pi}{2}\right) - 2 \tan\left(-\frac{\pi}{2}\right) = 4$	2

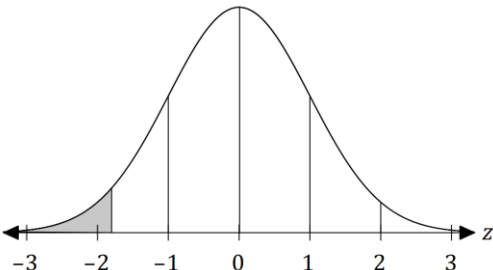
Question 18		
(a)	The population is Year 12 students in Australia. The sample is Year 12 students at her school.	1
(b)	Padma could obtain a list of the 330 students, pick a random starting number and then choose every 11 th person ($330 \div 30 = 11$) to be in her study.	2
(c)	(i) Padma has found a weak negative relationship between nicotine use and ATAR.	2
	When $x = 16$, y is approximately 52	3

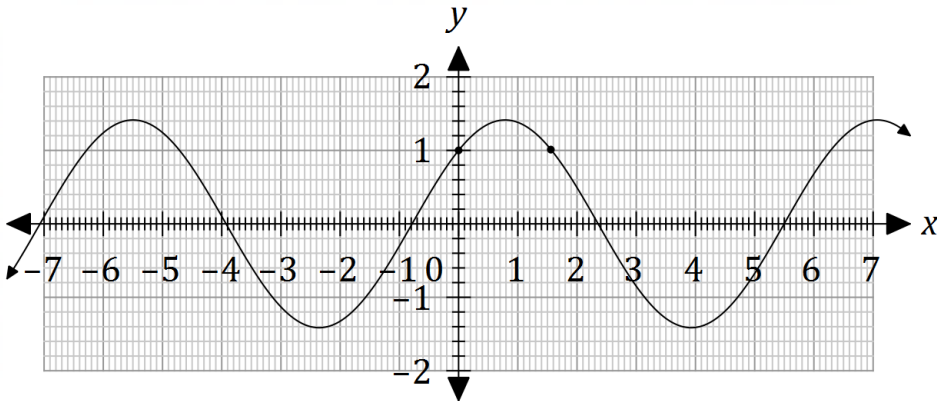
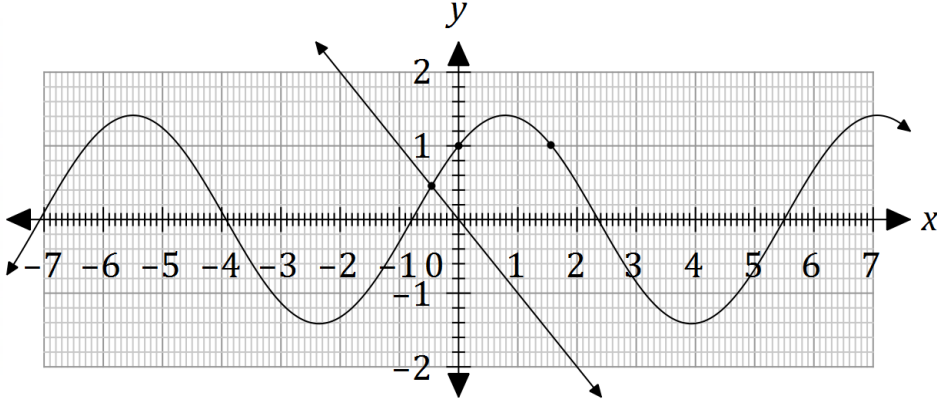
Question 18 (continued)			
	(iii)	Padma is incorrect. Two reasons such as those from the following list, may be presented: - Correlation does not imply causation - Padma only sampled students from her own school/from one specific cohort, which may not be representative of all Year 12 students in Australia - The correlation is weak - The sample size is small	3

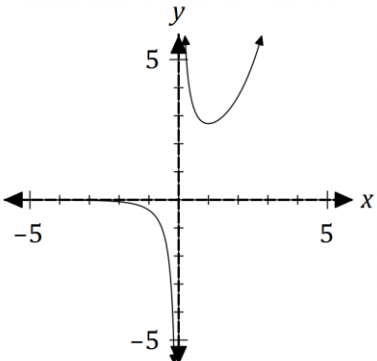
Question 19	
$y' = 4x^3 - 4x + 1$ $y'' = 12x^2 - 4$ $12x^2 - 4 = 0$ $x^2 = \frac{1}{3}$ $x = \pm \frac{1}{\sqrt{3}}$ When $x = -1, y'' = 8 > 0$ When $x = 0, y'' = -4 < 0$ When $x = 1, y'' = 8 > 0$  $(-\infty, -\frac{1}{\sqrt{3}}) \cup (\frac{1}{\sqrt{3}}, \infty)$ OR $x < -\frac{1}{\sqrt{3}}$ or $x > \frac{1}{\sqrt{3}}$	3

Question 20	
Events are independent if $P(B A) = P(B)$ $P(B A) = \frac{1}{6}$ $P(B) = \frac{6}{36} = \frac{1}{6}$ Therefore, these are independent events.	2

Question 21		
(a)	We cannot determine whether $\angle ACB$ is acute or obtuse, due to the length of BC being unknown. Thus, this is the ambiguous case.	1
(b)	Using sine rule to determine $\angle ACB$: $\theta = \sin^{-1}\left(\frac{803 \sin 63}{720}\right) = 83.58^\circ \text{ or } 96.42^\circ$ If $\angle ACB = 83.58^\circ, \angle CAB = 33.42^\circ$ $BC = \frac{720 \sin 33.42}{\sin 63} = 445.07 \text{ km}$ If $\angle ACB = 96.42^\circ, \angle CAB = 20.58^\circ$ $BC = \frac{720 \sin 20.58}{\sin 63} = 284.05 \text{ km}$	3

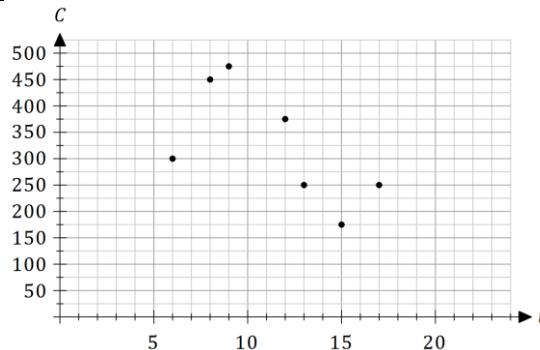
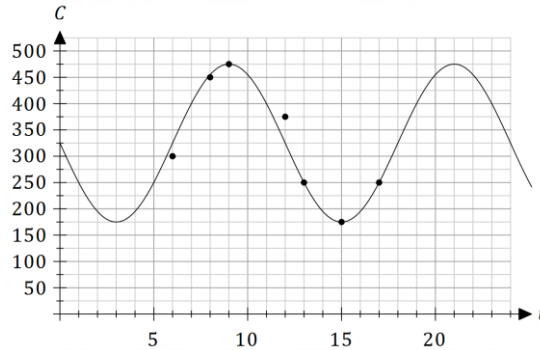
Question 22		
(a)		1
(b)	$z = \frac{x - \mu}{\sigma}$ $-1.8 = \frac{x - 7.5}{1.1}$ Multiply both sides by 1.1: $-1.98 = x - 7.5$ Add 7.5 to both sides: 5.5 lbs	2
(c)	Area to the right of $z = 1.8$ is $1 - 0.9641 = 0.0359$ $1000 \times 0.0359 = 35.9 \approx 36$	2

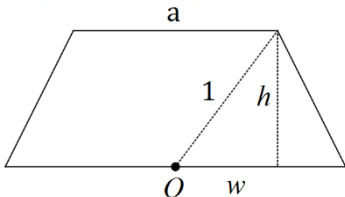
Question 23		
(a)	 $x \approx 0.16$	1
(b)	Rearranging the equation: $\sin(x) + \cos(x) = -x$ Graphing $y = -x$:  $x \approx -0.5$	2

Question 24		
(a)	x -intercepts occur at $y = 0$ $\frac{e^x}{x} = 0 \rightarrow e^x = 0$ There is no valid index that produces a result of 0. ($e^x > 0$ for all real values of x .) Hence, no x -intercepts.	1
(b)	$f'(x) = \frac{xe^x - e^x}{x^2} = 0$ $\frac{e^x(x-1)}{x^2} = 0$ $e^x(x-1) = 0$ $\therefore x = 1$ @ $x = 1, y = e$ $\therefore P(1, e)$ @ $x < 1, f'(x)$ is negative @ $x > 1, f'(x)$ is positive $\therefore P(1, e)$ is a local minimum	3
(c)	Vertical asymptote exists at $x = 0$ $f''(x) = \frac{x^3e^x - 2x^2e^x + 2xe^x}{x^4} = \frac{xe^x(x^2 - 2x + 2)}{x^4} = \frac{e^x(x^2 - 2x + 2)}{x^3}$ Point of inflection when $f''(x) = 0$ $\frac{e^x(x^2 - 2x + 2)}{x^3} = 0$ $e^x(x^2 - 2x + 2) = 0$ $x^2 - 2x + 2 = 0$ $\Delta = 4 - 4(2)(1) = -4$ Discriminant < 0 results in no real roots – hence, $f''(x) \neq 0$ and no point of inflection. @ $x < 0, f''(x) < 0$ – thus, concave down when $x < 0$ @ $x > 0, f''(x) > 0$ – thus, concave up when $x > 0$	4
(d)		2

Question 25		
(a)	Area under $f(x)$ on the given domain must be equal to 1. $\int_0^4 (kx^2 - 3kx + 4k)dx = 1$ $k \left[\frac{x^3}{3} - \frac{3x^2}{2} + 4x \right]_0^4 = 1$ $k \left[\frac{64}{3} - \frac{3(16)}{2} + 16 \right] - [0] = 1$ $k \left(\frac{40}{3} \right) = 1$ $k = \frac{3}{40}$	2
(b)	The mode is the global max on the given domain. The PDF is a concave-up parabola, so there is no local maximum. Testing endpoints: $f(0) = \frac{3}{10}, f(4) = 1$ The highest point on the distribution is at $x = 4$, hence this is the mode. The least likely outcomes are values close to the lowest point on the distribution. The vertex is a global min, which occurs at $x = -\frac{b}{2a} = -\frac{-\frac{9}{40}}{2\left(\frac{3}{40}\right)} = \frac{3}{2}$. Hence, the least likely outcomes are those close to $\frac{3}{2}$.	2
(c)	$P(X < 3) X > 2.77 = \frac{P(X < 3 \cap X > 2.77)}{P(X > 2.77)}$ Using working from part (a), the CDF of X is $F(x) = \frac{3}{40} \left(\frac{x^3}{3} - \frac{3x^2}{2} + 4x \right)$ and $P(X < 3) = \frac{3}{40} \left(\frac{27}{3} - \frac{3(9)}{2} + 12 \right) = \frac{9}{16}$ (another integral may also be used to find this) $P(X < 3 \cap X > 2.77) = P(X < 3) - 0.5 = \frac{9}{16} - \frac{1}{2} = \frac{1}{16}$ $P(X < 3) X > 2.77 = \frac{\frac{1}{16}}{\frac{1}{2}} = \frac{1}{8}$	2

Question 26		
(a)	$A_0 = 580000, r = 0.042 \div 12 = 0.0035$ $A_n = A_{n-1}(1.0035) - M$ $A_1 = 580000(1.0035) - M$ $A_2 = [580000(1.0035) - M](1.0035) - M$ $\quad = 580000(1.0035)^2 - (1.0035)M - M$ $A_3 = [580000(1.0035)^2 - (1.0035)M - M](1.0035) - M$ $\quad = 580000(1.0035)^3 - (1.0035)^2M - (1.0035)M - M$ The last three terms are a geometric series with $a = -M$, $r = 1.0035$, and n terms. $A_n = 580000(1.0035)^n - \frac{M(1.0035^n - 1)}{0.0035}$	2
(b)	$A_n = 0$ when $n = 30 \times 12 = 360$ $0 = 580000(1.0035)^{360} - \frac{M(1.0035^{360} - 1)}{0.0035}$ $M = 580000(1.0035)^{360} \div \frac{(1.0035^{360} - 1)}{0.0035} \approx \2836.30 (nearest cent)	2
(c)	Months to pay off loan if $M = 2936.30$: $0 = 580000(1.0035)^n - \frac{2936.30(1.0035^n - 1)}{0.0035}$ $0 = 580000(1.0035)^n - \frac{2936.30(1.0035)^n}{0.0035} + \frac{2936.30}{0.0035}$ $0 = -258942.8571(1.0035)^n + 838942.8571$ $(1.0035)^n = 3.2398 \dots$ $n \ln 1.0035 = \ln 3.2398$ $n \approx 336$ Total paid without extra payments: $2836.3 \times 360 = 1021068$ Total paid with extra payments: $2936.3 \times 336 = 986596.8$ Saved: $1021068 - 986596.8 = \\$34,500$ (nearest \$100)	4

Question 27		
(a)		1
(b)	Find A: Maximum = 475, Minimum = 175 $A = \frac{475-175}{2} = 150$ Find D: Halfway between max and min. $D = \frac{475+175}{2} = 325$ Find B: Period is 12 hours. $B = \frac{2\pi}{12} = \frac{\pi}{6}$ Find C: First max of sine is normally at $\theta = \frac{\pi}{2}$. In the data it occurs at $t = 9$. $\frac{\pi}{6}(9) + C = \frac{\pi}{2} \rightarrow C = -\pi$ $C(t) = 150 \sin\left(\frac{\pi}{6}t - \pi\right) + 325$	4
(c)		1
(d)	$C(20) = 150 \sin\left(\frac{\pi}{6}(20) - \pi\right) + 325 \approx 455 \text{ pmol/L}$ $C'(t) = 25\pi \cos\left(\frac{\pi}{6}t - \pi\right)$ $C'(20) = 25\pi \cos\left(\frac{\pi}{6}(20) - \pi\right) \approx 39 \text{ pmol/L per hour}$	2
(e)	$200 = 150 \sin\left(\frac{\pi}{6}t - \pi\right) + 325$ $\sin\left(\frac{\pi}{6}t - \pi\right) = -\frac{5}{6}$ $\frac{\pi}{6}t - \pi = \sin^{-1}\left(-\frac{5}{6}\right)$ or $\frac{\pi}{6}t - \pi = \pi - \sin^{-1}\left(-\frac{5}{6}\right)$ $t \approx 4.119 \dots$ or $13.881 \dots = 4 \text{ h } 7 \text{ m}$ or $13 \text{ h } 53 \text{ m}$ The next times can be found by adding the period 12 hours: $16 \text{ h } 7 \text{ m}$, $25 \text{ h } 53 \text{ m}$	3

Question 28		
(a)	Find the length of the base which forms a chord of the circle in terms of h:  $w = \sqrt{1 - h^2} \text{ (Pythagoras)}$ $a = 2w = 2\sqrt{1 - h^2}$ $A = \frac{h}{2}(a + b) = \frac{h}{2}(2\sqrt{1 - h^2} + 2)$ $= h\sqrt{1 - h^2} + h$	2
(b)	$A' = \sqrt{1 - h^2} + \frac{h}{2} \times \frac{-2h}{\sqrt{1 - h^2}} + 1$ $= \sqrt{1 - h^2} + 1 - \frac{h^2}{\sqrt{1 - h^2}}$ $= \frac{\sqrt{1 - h^2}(\sqrt{1 - h^2} + 1) - h^2}{\sqrt{1 - h^2}}$ $= \frac{(1 - h^2) + \sqrt{1 - h^2} - h^2}{\sqrt{1 - h^2}}$ $= \frac{1 - 2h^2 + \sqrt{1 - h^2}}{\sqrt{1 - h^2}} = 0$ $1 - 2h^2 + \sqrt{1 - h^2} = 0$ (the fraction will = 0 when the numerator = 0) $\sqrt{1 - h^2} = 2h^2 - 1$ (by subtracting $(1 - 2h^2)$ from both sides) $1 - h^2 = 4h^4 - 4h^2 + 1$ (by squaring both sides) $4h^4 - 3h^2 = 0$ (by moving all terms to RHS) $h^2(4h^2 - 3) = 0$ (by factorizing) $h = 0 \text{ or } h = \frac{\sqrt{3}}{2}$ When $h = 0.5$, $A' = 1.57 \dots > 0$ and when $h = 0.9$, $A' = -0.42 < 0$, hence $h = \frac{\sqrt{3}}{2}$ gives a maximum. Substitute h into formula for A : $A_{\max} = \frac{\sqrt{3}}{2} \sqrt{1 - \left(\frac{\sqrt{3}}{2}\right)^2} + \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{4}$	4